WORKED SOLUTION

STEP III 2022, Question 6

Pure • Rolling Circles and Tangents

STEP III 2022, Question 6

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

The key idea

Part (i) is a second-order Taylor expansion, and its real point is the warning in the last line: when the first-order terms vanish, the ratio is decided by the second-order terms. In part (ii) the rolling condition says that equal arcs of the two circles pass the point of contact, which tells us how far C_1 has turned. At the special point in (c) the first-order terms in the chord cancel exactly, so we need the second-order terms from (i).

Solution

Part (i)

For small α, $\cos\alpha \approx 1 - \frac{1}{2}\alpha^2$ and $\sin\alpha \approx \alpha$, with errors of order α^3. So

$$\begin{aligned}\cos(\theta+\alpha) - \cos\theta &= \cos\theta\cos\alpha - \sin\theta\sin\alpha - \cos\theta \\ &\approx \cos\theta(1 - \tfrac{1}{2}\alpha^2) - \alpha\sin\theta - \cos\theta = -\alpha\sin\theta - \tfrac{1}{2}\alpha^2\cos\theta,\end{aligned}$$

as required. In the same way,

$$\sin(\theta+\alpha) - \sin\theta \approx \alpha\cos\theta - \tfrac{1}{2}\alpha^2\sin\theta.$$

Dividing top and bottom of $(*)$ by α,

$$\frac{\sin(\theta+\alpha)-\sin\theta}{\cos(\theta+\alpha)-\cos\theta} \approx \frac{\cos\theta - \frac{1}{2}\alpha\sin\theta}{-\sin\theta - \frac{1}{2}\alpha\cos\theta} \longrightarrow -\frac{\cos\theta}{\sin\theta} = -\cot\theta,$$

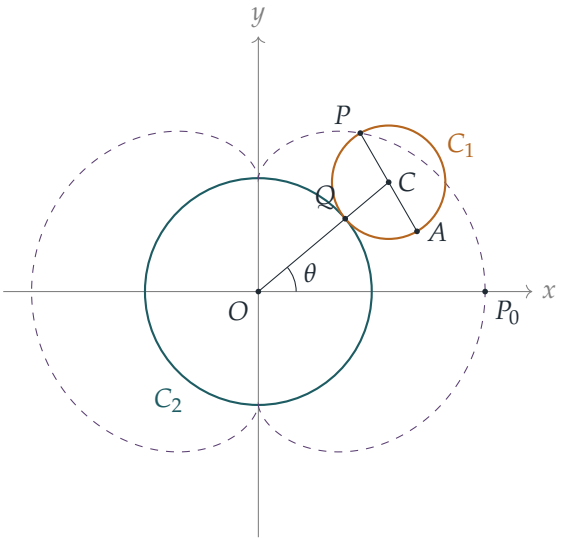
and the limit is fine because the denominator tends to $-\sin\theta \neq 0$.

If $\sin\theta = 0$ then $\cos\theta = \pm 1$, and the first-order term of the denominator disappears:

$$(*) \approx \frac{\alpha\cos\theta}{-\frac{1}{2}\alpha^2\cos\theta} = -\frac{2}{\alpha}.$$

So $(*)$ has no finite limit: its size grows without bound, tending to $-\infty$ as $\alpha \to 0^+$ and to $+\infty$ as $\alpha \to 0^-$. (Geometrically, the chord of the unit circle from angle θ to $\theta+\alpha$ becomes vertical, and its gradient blows up.)

Part (ii)(a)



The figure shows $n = 3$, with the dashed curve the path of P. Let C be the centre of C_1. Since the circles touch at Q, the points O, Q, C are collinear with $OC = (n-1)a + a = na$, so

$$\overrightarrow{OC} = na(\cos\theta, \sin\theta),$$

and from C the point Q is in direction $\theta + \pi$.

Let A be the point of C_1 diametrically opposite P; this is the point of C_1 that was touching C_2 at the start, at $((n-1)a, 0)$. Rolling without slipping means the arc of C_1 from A to Q equals the arc of C_2 that the contact has travelled, namely $(n-1)a\,\theta$. Since C_1 has radius a, the angle ACQ is $(n-1)\theta$.

Ask yourself Is A clockwise or anticlockwise from Q, as seen from C?

Rolling anticlockwise round the outside of C_2, the circle C_1 spins anticlockwise, so the points it has already rolled over are carried anticlockwise away from the contact point. So $\overrightarrow{CA}$ points in direction $\theta + \pi + (n-1)\theta = n\theta + \pi$, and $\overrightarrow{CP} = -\overrightarrow{CA}$ points in direction $n\theta$. (Check: at $\theta = 0$ this gives $P = ((n+1)a, 0)$. Altogether C_1 has turned through $n\theta$, the familiar "one extra turn" of a coin rolled round another.) Hence

$$\overrightarrow{OP} = na(\cos\theta, \sin\theta) + a(\cos n\theta, \sin n\theta),$$

so $x(\theta) = a(n\cos\theta + \cos n\theta)$, as required, and

$$y(\theta) = a(n\sin\theta + \sin n\theta).$$

Part (ii)(b)

Squaring and adding,

$$OP^2 = a^2(n^2 + 1 + 2n(\cos\theta\cos n\theta + \sin\theta\sin n\theta)) = a^2(n^2 + 1 + 2n\cos(n-1)\theta).$$

Setting this equal to $(n-1)^2a^2 = (n^2-2n+1)a^2$ gives $2n\cos(n-1)\theta = -2n$, that is $\cos(n-1)\theta = -1$. So

$$\theta = \frac{(2k+1)\pi}{n-1}, \qquad k \in \mathbb{Z},$$

giving $n-1$ distinct positions in each turn round C_2. These are the moments when P itself is the point of contact: the cusps of the curve.

Part (ii)(c)

With $\theta_0 = \pi/(n-1)$ we have $n\theta_0 = \theta_0 + \pi$, so

$$\cos n\theta_0 = -\cos\theta_0, \qquad \sin n\theta_0 = -\sin\theta_0.$$

Apply part (i) to each term, using $n\alpha$ as the small increment for the $n\theta$ terms:

$$\begin{aligned}
\frac{y(\theta_0+\alpha)-y(\theta_0)}{a} &\approx n(\alpha\cos\theta_0 - \tfrac{1}{2}\alpha^2\sin\theta_0) \\
&\quad + (n\alpha\cos n\theta_0 - \tfrac{1}{2}n^2\alpha^2\sin n\theta_0) \\
&= n\alpha\cos\theta_0 - \tfrac{1}{2}n\alpha^2\sin\theta_0 - n\alpha\cos\theta_0 + \tfrac{1}{2}n^2\alpha^2\sin\theta_0 \\
&= \tfrac{1}{2}n(n-1)\alpha^2\sin\theta_0, \\
\frac{x(\theta_0+\alpha)-x(\theta_0)}{a} &\approx n(-\alpha\sin\theta_0 - \tfrac{1}{2}\alpha^2\cos\theta_0) \\
&\quad + (-n\alpha\sin n\theta_0 - \tfrac{1}{2}n^2\alpha^2\cos n\theta_0) \\
&= \tfrac{1}{2}n(n-1)\alpha^2\cos\theta_0,
\end{aligned}$$

each with an error of order α^3. The first-order terms cancel exactly, just as in the $\sin\theta = 0$ case of (i), so the second-order terms decide the answer. Dividing top and bottom by $\frac{1}{2}n(n-1)\alpha^2$,

$$\frac{y(\theta_0+\alpha)-y(\theta_0)}{x(\theta_0+\alpha)-x(\theta_0)} = \frac{\sin\theta_0 + O(\alpha)}{\cos\theta_0 + O(\alpha)} \longrightarrow \tan\theta_0,$$

provided $\cos\theta_0 \neq 0$, which holds for $n \geq 4$. When $n = 3$, $\theta_0 = \frac{1}{2}\pi$; the same working shows the reciprocal ratio tends to $\cot\theta_0 = 0$, so the chord becomes vertical.

This limit is the gradient of the tangent at θ_0, because it is the limit of chord gradients. Now

$$x(\theta_0) = a(n\cos\theta_0 - \cos\theta_0) = (n-1)a\cos\theta_0, \qquad y(\theta_0) = (n-1)a\sin\theta_0,$$

so OP makes angle θ_0 with the x-axis and has gradient $\tan\theta_0$ (it is vertical when $n = 3$). The tangent and OP have the same direction, so the tangent at $(x(\theta_0), y(\theta_0))$ is parallel to OP. In fact it lies along the line OP, since it passes through P.

Checks

- For $\theta = 0$, $\alpha = 0.001$ the ratio $(*)$ is -1999.9998, against $-2/\alpha = -2000$.

- With $n = 5$, $\theta_0 = \frac{1}{4}\pi$ and $\alpha = 10^{-4}$, the chord gradient is 1.0004, close to $\tan\frac{1}{4}\pi = 1$.
- At θ_0, $OP = (n-1)a$, so the point of (c) is the $k = 0$ cusp found in (b). That is why the tangent there is radial.

Common slips

- Taking the rotation of C_1 to be $(n-1)\theta$ (the arc rolled) and forgetting the extra θ from C travelling round O.
- Using only first-order approximations in (c): they give $0/0$, which tells you nothing.
- Forgetting that $n = 3$ gives $\theta_0 = \frac{1}{2}\pi$, where $\tan\theta_0$ is undefined.

Beyond STEP

The curve is an epicycloid with $n-1$ cusps. At each cusp the velocity $(x'(\theta), y'(\theta))$ is zero, so the ordinary formula $\mathrm{d}y/\mathrm{d}x = y'/x'$ fails. Part (c) handles this by going to second order, which is l'Hôpital's rule in disguise. Points where a parametrised curve has zero velocity are called singular points, and they are studied in a first course on the differential geometry of curves. For the limits and Taylor approximations behind part (i), see Abbott, *Understanding Analysis*.

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